

WASTE AND CONTAMINATED-LAND HAZARD

Waste / Dump Sites

Modern landfills are engineered facilities, while old dumps and buried fill may have uncertain contents, gas, leachate, settlement, and fire behavior.



1 Understand the factor

What it is - and why it matters

Waste sites include municipal landfills, construction-debris fills, transfer stations, recycling operations, composting sites, illegal dumps, closed burn dumps, and historic disposal areas later redeveloped. Decomposing organic waste can generate methane, carbon dioxide, odors, heat, and leachate. Buried waste can settle for decades and may contain hazardous materials, ash, drums, or construction debris. Operational impacts can include trucks, dust, birds, pests, noise, litter, and nighttime lighting.

How risk can reach a household

- Landfill gas can migrate through soil and utility trenches, accumulate in enclosed spaces, and create explosion or asphyxiation hazards.
- Leachate and stormwater can carry contaminants into groundwater, streams, or neighboring soil when collection systems fail or were never installed.
- Subsurface heating, chemical reactions, and landfill fires can release smoke, odors, and irritating compounds for extended periods.
- Uneven settlement can damage foundations, roads, utilities, drainage, caps, and structures built over or near waste.

What people often miss

- A closed landfill may be beneath a park, subdivision, shopping center, or vacant lot and may not look like a dump.
- A landfill-gas collection system controls but does not eliminate gas generation; wells, flares, and monitoring need maintenance.
- Complaints about odor do not by themselves identify toxicity, but recurring odor can signal operational problems and affect quality of life.
- Birds, vectors, windblown litter, trucks, and fire risk can extend beyond mapped waste boundaries.

? Five checks before buying

1

Search state and local solid-waste, cleanup, landfill, fire, air, water, and code-enforcement records, including former names and parcel history.

2

Review waste types, operating status, closure plans, gas and leachate systems, monitoring data, violations, fires, odors, and groundwater flow.

3

Check title and planning records for deed restrictions, methane zones, special foundations, vapor barriers, soil caps, and excavation controls.

4

Use geotechnical and environmental professionals when the parcel contains fill, is near a gas plume, or has unexplained settlement or odors.

5

Ask the insurer about pollution, smoke, earth movement, settling, sewer backup, remediation systems, and loss-of-use coverage.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 13

H Health impacts

- Smoke or high gas concentrations can cause headache, nausea, eye and airway irritation, breathing difficulty, or acute danger in confined spaces.
- Pests, litter, truck emissions, noise, and nighttime operations can affect outdoor activity, sleep, stress, and respiratory comfort.
- Chronic health concerns depend on specific measured chemicals and exposure; odors alone cannot establish dose or disease.
- Children may have higher contact with contaminated soil or dust if caps are thin, damaged, or disturbed.

P Property impacts

- Settlement and buried debris can crack slabs, foundations, roads, pools, pipes, sewers, drainage systems, and landscaping.
- Smoke, odor, traffic, litter, vectors, and visual impacts can reduce enjoyment and affect buyer perception or resale.
- Gas mitigation, specialized foundations, monitoring, cap maintenance, and soil-management plans can add long-term cost.
- A facility emergency or access-road closure can disrupt nearby neighborhoods even without direct property damage.

I Insurance implications

- Homeowners policies often exclude pollution, land, gradual seepage, settling, and pre-existing waste conditions.
- Required methane systems and monitoring may not be insured against maintenance failure or code-driven replacement.
- Smoke or fire damage may be covered when sudden and direct, but odor, diminished value, preventive relocation, and testing may be limited.
- Environmental liability may be complicated when waste lies on the parcel, so legal and environmental due diligence are important.

Possible long-term health implications of buying nearby

- Long-term health impact depends on whether gas, smoke, contaminated soil, dust, or water reaches the household at meaningful concentrations.
- Repeated odor and smoke events can create headaches, anxiety, sleep loss, and reduced outdoor activity even when toxicological significance is uncertain.
- A properly closed and monitored landfill can differ substantially from an undocumented dump with uncontrolled waste and settlement.
- Changes in drainage, wildfire, excavation, or gas-system operation can alter conditions, so continued monitoring matters.

Evidence lens

Landfill gas, settlement, leachate, and fire mechanisms are established. Health assessment requires compound-specific monitoring and a confirmed pathway rather than distance or odor alone.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 13

V Verify the actual risk

- Obtain operating or closure plans, waste footprint maps, gas and groundwater data, enforcement history, and complaint logs.
- Confirm whether the home sits on native soil, engineered fill, refuse, a gas-migration zone, or a deed-restricted parcel.

P Protect the property

- Maintain required gas barriers, vents, alarms, caps, drainage, and settlement-tolerant utilities; never block monitoring points.
- Avoid excavation or deep planting in capped or restricted areas without approval and a soil-management plan.

M Monitor and respond

- Use combustible-gas or indoor-air monitoring when required; report persistent odor, smoke, heat, settlement, or dead vegetation.
- During a fire or air advisory, close openings, use recirculated filtered air, and follow evacuation or shelter instructions.

I Prepare insurance records

- Clarify pollution, smoke, fire, earth-movement, settling, testing, and additional-living-expense coverage.
- Keep monitoring records, system maintenance, engineering reports, photographs, and agency correspondence.

REAL-LIFE EXAMPLE

Chiquita Canyon Landfill Subsurface Reaction

2022-present response

A subsurface reaction at the Chiquita Canyon Landfill in Los Angeles County produced elevated temperatures, landfill gas, leachate, odors, and community complaints. EPA and state and local agencies issued orders and conducted monitoring and response work, while nearby residents reported significant quality-of-life impacts.

Why it matters: The case demonstrates that landfill conditions can change after years of operation and that heat, gas, leachate, odor, monitoring, and agency coordination can become long-duration community issues.

[Open official incident record - epa.gov >](#)

Questions to resolve before making a decision

1. Search state and local solid-waste, cleanup, landfill, fire, air, water, and code-enforcement records, including former names and parcel history.
2. Review waste types, operating status, closure plans, gas and leachate systems, monitoring data, violations, fires, odors, and groundwater flow.
3. Check title and planning records for deed restrictions, methane zones, special foundations, vapor barriers, soil caps, and excavation controls.
4. Use geotechnical and environmental professionals when the parcel contains fill, is near a gas plume, or has unexplained settlement or odors.

Official sources and mapping tools

[EPA endangerment finding - epa.gov >](#)[EPA landfill basics - epa.gov >](#)[EPA Landfill Methane Outreach Program - epa.gov >](#)